CNN VAD Diagnostic Report

1 Diagnostic Analysis: Why CNN VAD Fails on Real Audio

1.1 Problem Observation

We evaluated both an energy-based VAD baseline and a CNN-based VAD model on real audio (`sample.wav`).

The energy-based method performs reasonably well:

- F1 score ≈ 0.92 (after smoothing)
- Low false alarm rate

However, the CNN model fails completely:

- F1 score $= 0.4047$
- False Alarm Rate $= 1.0$
- All frames are predicted as speech

This indicates a severe generalization issue.

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1.2 Initial Debugging Attempts

We tested several possible causes:

- Frame alignment mismatch $\rightarrow$ fixed, no improvement
- Threshold selection $\rightarrow$ no effect (all outputs > 0.9)
- Feature normalization $\rightarrow$ no effect

We observed that the CNN outputs are saturated:

- Mean probability ≈ 0.999
- All frames classified as speech

This suggests that the model has collapsed to a degenerate solution.

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1.3 Feature Distribution Analysis

To further diagnose the issue, we compared feature distributions between synthetic training data and real audio.

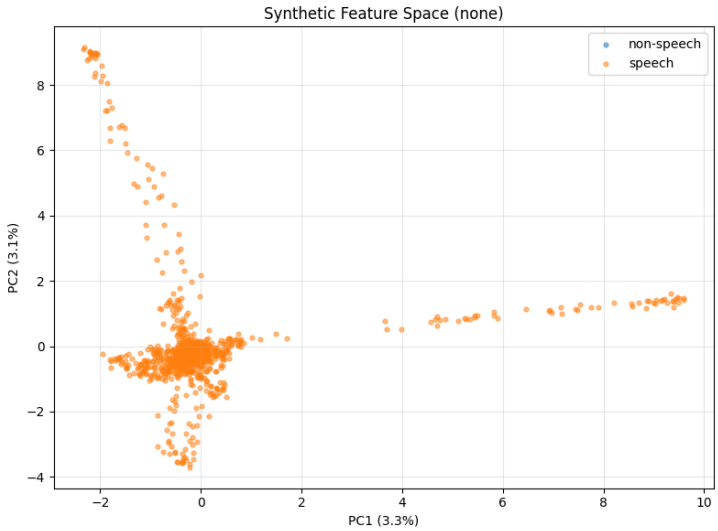

Figure 1: PCA visualization of synthetic dataset

1.3.1 Synthetic Feature Space

Observations:

- Highly structured, cross-like distribution
- Most samples concentrated near the origin
- No dominant principal direction (PC1 $\approx$ 3% variance)

This indicates that the synthetic data lies on a highly constrained and artificial manifold.
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1.3.2 Real Audio Feature Space

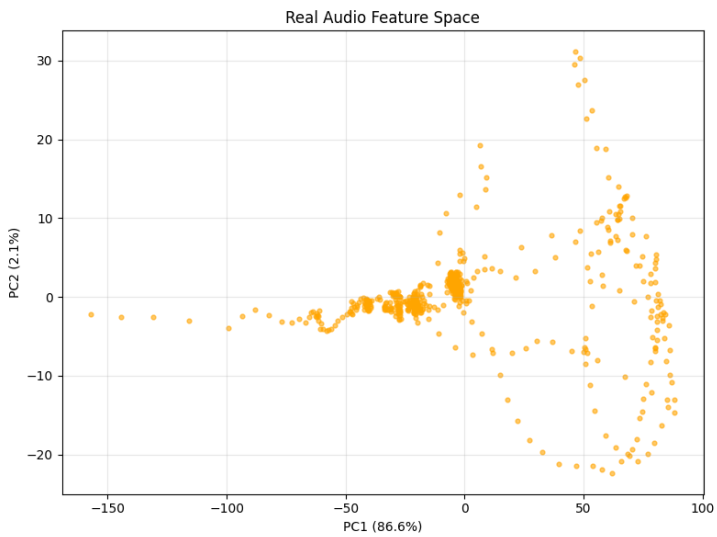

Figure 2: PCA visualization of real audio features

Observations:

- Large variance and irregular structure
- Continuous distribution rather than discrete clusters
- Strong principal direction (PC1 $\approx$ 86.6% variance)

This reflects the complexity of real-world acoustic signals.

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1.3.3 Direct Comparison

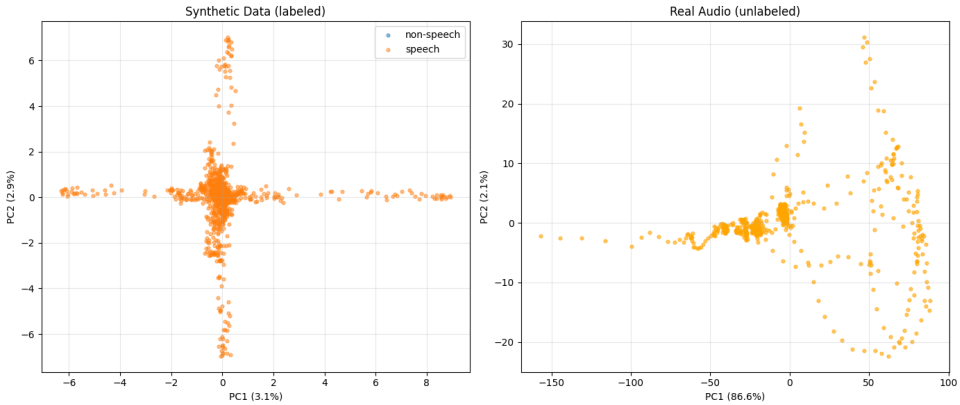

Figure 3: Side-by-side comparison of synthetic vs real feature spaces

Key difference:

- Synthetic data: low-variance, structured, artificial geometry
- Real data: high-variance, continuous, natural geometry

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1.4 Feature Value Distribution

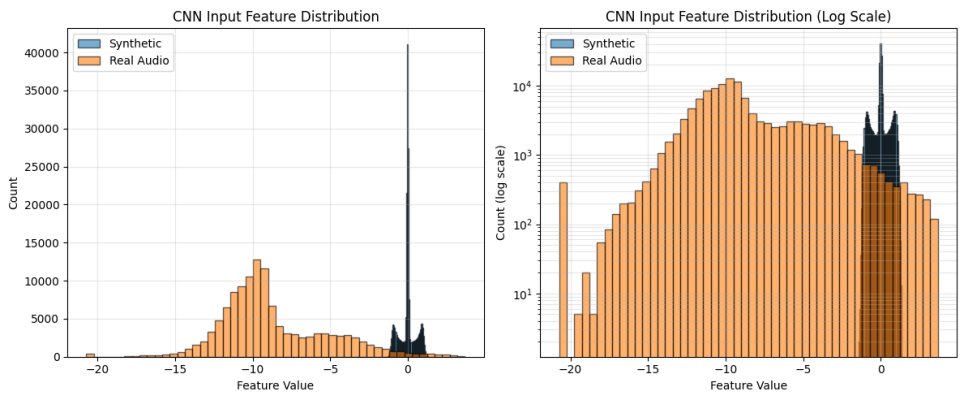

Figure 4: Feature value distribution: synthetic vs real

Observations:

- Synthetic features: centered around 0, narrow range

- Real features: large negative shift, much wider spread

Even after normalization, this mismatch persists in structure.

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1.5 CNN Output Behavior

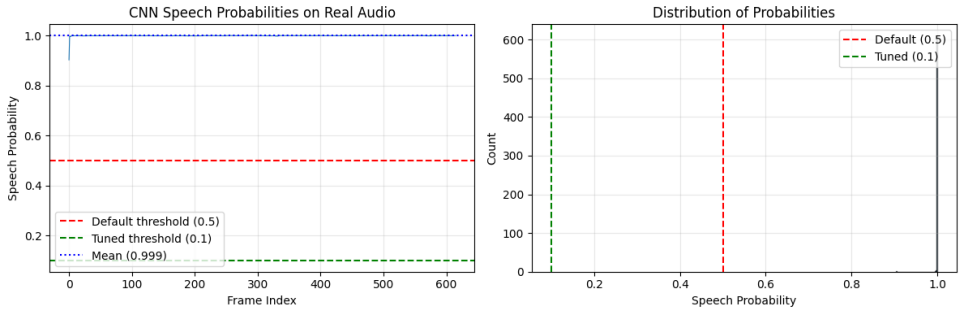

Figure 5: CNN output probabilities on real audio

Observations:

- All probabilities $\in [0.9, 1.0]$
- No separation between speech and non-speech

This explains why threshold tuning has no effect.

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1.6 Comparison with Energy-based VAD

The energy-based method works because:

- It relies on a simple, robust feature (log energy)
- It does not depend on learned feature geometry

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1.7 Root Cause

The failure is caused by **severe domain shift**:

- Synthetic data does not resemble real-world audio
- The model learns shortcut patterns specific to synthetic data
- These patterns do not generalize

As a result:

> All real audio inputs fall into the model's learned "speech region".

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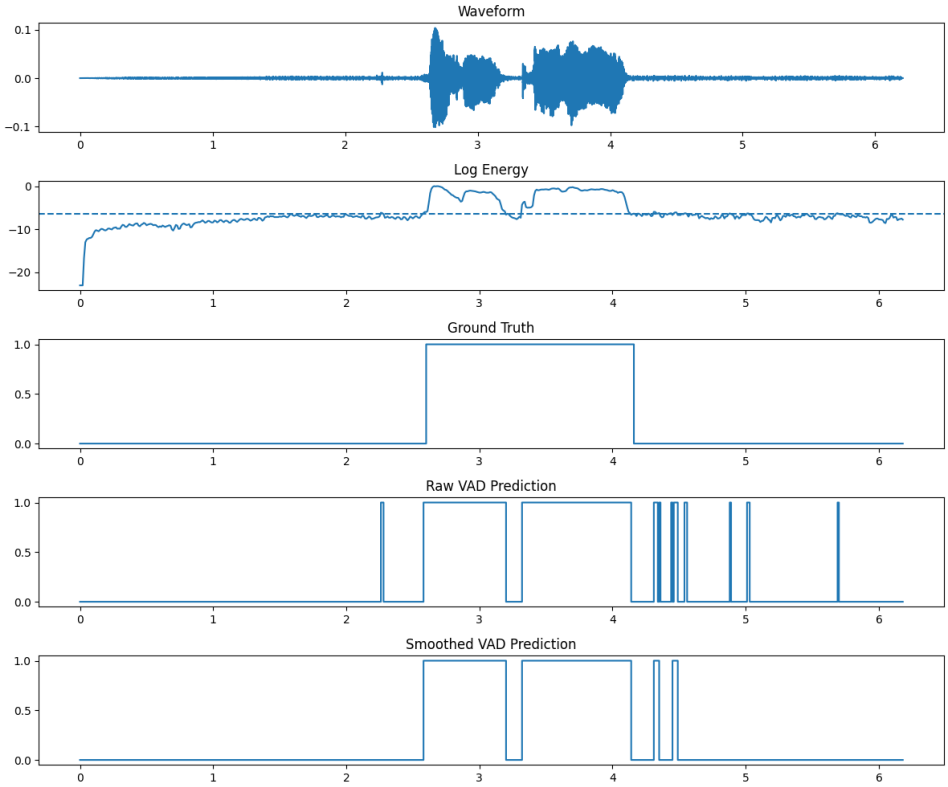

Figure 6: Energy-based VAD vs ground truth

1.8 Conclusion: Necessity of Changing Training Data

Based on the above analysis, we conclude:

- The issue is not due to implementation errors
- The issue is not due to threshold or normalization
- The core problem is unrealistic training data

Therefore:

> **Improving or replacing the training dataset is necessary for the CNN model to generalize.**

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1.9 Next Step

We plan to:

- Replace synthetic data with real datasets (e.g., LibriSpeech, MUSAN)
- Or augment synthetic data to better match real-world feature distributions